

Cybersecurity Training Diary

Day 11: Introduction to Ethical Hacking & Kali Linux

1. Topics Covered

✓ Ethical Hacking Fundamentals

Term	Definition	Example
White Hat	Legal security testing	Penetration tester
Black Hat	Malicious hacking	Cybercriminal
Bug Bounty	Rewards for vulnerabilities	HackerOne programs

✓ Kali Linux Tools Overview

```
pie
  title Top Kali Tools
  "Nmap" : 25
  "Metasploit" : 20
  "Burp Suite" : 15
  "Wireshark" : 15
  "John the Ripper" : 10
  "Others" : 15
```

✓ Legal Considerations

- **Written permission** required for testing
- **Scope definition:** What systems to test
- **Disclosure policies:** How to report findings

2. Key Takeaways

Authorization is everything - unauthorized testing = crime

Document all actions for penetration test reports

Kali = Swiss Army knife for security pros

3. Practical Exercises

Exercise 1: Kali Linux Setup

1. Download Kali ISO from kali.org
2. Create bootable USB with [BalenaEtcher](https://balena.io/etcher)
3. Boot in "Live" mode (no installation needed)

Exercise 2: First Nmap Scan

```
nmap -sV -O 192.168.1.1  
# -sV: Service detection  
# -O: OS fingerprinting
```

4. Questions for Reflection

What's the difference between vulnerability scanning and penetration testing?

Why does Kali Linux come with so many hacking tools pre-installed?

Name - Bhawanpreet Kaur

URN - 2302499